

Note: Please answer all the questions in English. Your grade is determined by the minimum of the points you obtained and 40.

Problem 1. (6 points)

Let X and Y be independent binomial variables, $\text{bin}(m, p)$ and $\text{bin}(n, p)$, respectively. Find the distribution of $X + Y$.

Problem 2. (6 points)

Let Y be distributed as $\text{bin}(n, X)$, where X is a random variable uniformly distributed on $[0, 1]$.

- a) Find $\mathbf{E}[Y]$.
- b) Find $\text{var}(Y)$.

Problem 3. (6 points)

In your pocket is a random number N of coins, where N has the Poisson distribution with parameter λ . You toss each coin once, with heads showing with probability p each time. We denote the total number of heads as X .

- a) Find the transform of X .
- b) Find $\text{var}(X)$.

Problem 4. (6 points)

Let X and Y be independent random variables each having the uniform distribution on $[0, 1]$. Let $U = \min\{X, Y\}$ and $V = \max\{X, Y\}$.

- a) Find $\mathbf{E}[U]$.
- b) Find $\text{cov}(U, V)$.

Problem 5. (7 points)

Let $X_1, X_2, \dots, X_n$ be independent random variables. For each $i \in \{1, 2, \dots, n\}$, X_i follows a **standard normal distribution**. Define $Y = X_1^2 + X_2^2 + \dots + X_n^2$.

- a) Find $\mathbf{E}[Y]$ and $\text{var}(Y)$.
- b) Suppose that $n = 100$. Find a good approximation for the probability that $\mathbf{P}(Y \geq 400)$. Give your answer in terms of the standard normal CDF $\Phi(x)$.

Problem 6. (8 points)

A store opens at $t = 0$ and potential customers arrive in a Poisson manner at an average arrival rate of λ potential customers per hour. As long as the store is open, and independently of all other events, each particular potential customer becomes an actual customer with probability p . The store closes as soon as ten actual customers have arrived.

- a) What is the probability that the fifth potential customer to arrive becomes the third actual customer?
- b) Find the process for the arrival of actual customers and its parameter.
- c) What is the PDF and expected value for L , the duration of the interval from store opening to store closing?
- d) Considering ten actual customers have arrived, what is the probability that no two actual customers arrive within τ time units of each other?

Problem 7. (6 points)

A fair coin is tossed n times, showing heads H_n times and tails T_n times. Let $S_n = H_n - T_n$.

- a) Find the transform of S_n .
- b) Prove that

$$\mathbf{P}(S_n > an)^{1/n} \leq 1/\sqrt{(1+a)^{1+a}(1-a)^{1-a}}, \quad \text{if } 0 < a < 1.$$

Hint: Chernoff inequality.